

AI ASSISTED LAB_ASSIGNMENT_6.1

NAME:-k.vedhamsh

ROLL.NO:-2503A52708

BATCH:-24

Task Description #1 (AI-Based Code Completion for Loops)

Task: Use an AI code completion tool to generate a loop-based program.

Expected Output:

- AI-generated loop logic.

- Identification of loop type used (for or while).

- Validation with sample inputs.

Prompt:-

Generate Python code to print all even numbers between 1 and N using a loop

CODE:-

```
# Generate Python code to print all even numbers between 1 and N using a loop
```

```
N = int(input("Enter a number: "))
```

```
for i in range(1, N + 1):
```

```
    if i % 2 == 0:
```

```
        print(i)
```

OUTPUT:-

```
PS C:\Users\kisho_gtbim4w\OneDrive\WTMP LAB> & C:
```

```
\Users\kisho_gtbim4w\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/
```

```
Users/kisho_gtbim4w/OneDrive/WTMP LAB/AI.PYTHON"
```

```
Enter a number: 23
```

```
2
```

```
4
```

```
6
```

```
8
```

```
10
```

```
12
```

```
14
```

16

18

20

22

PS C:\Users\kisho_gtbim4w\OneDrive\WTMP LAB> & C:

\Users\kisho_gtbim4w\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/

Users/kisho_gtbim4w/OneDrive/WTMP LAB/AI.PYTHON"

Enter a number: 100

2

4

6

8

10

12

14

16

18

20

22

24

26

28

30

32

34

36

38

40

42

44

46

48

50

52

54

56

58

60

62

64

66

68

70

72

74

76

78

80

82
84
86
88
90
92
94
96
98
100

Task Description #2 (AI-Based Code Completion for Loop with Conditionals)

Task: Use an AI code completion tool to combine loops and conditionals.

Week3 -

Monday

Expected Output:

AI-generated code using loop and if condition.

Correct count validation.

Explanation of logic flow.

Prompt:

Generate Python code to count how many numbers in a list are even and odd

CODE:-

```
# Count how many numbers in a list are even and odd
numbers = [12, 5, 8, 17, 10, 3, 6, 9]
```

```
even_count = 0
odd_count = 0
```

```
for num in numbers:
    if num % 2 == 0:
        even_count += 1
    else:
        odd_count += 1
```

```
print("Even numbers:", even_count)
print("Odd numbers:", odd_count)
```

OUTPUT:-

```
PS C:\Users\kisho_gtbim4w\OneDrive\WTMP LAB> & C:
\Users\kisho_gtbim4w\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/
Users/kisho_gtbim4w/OneDrive/WTMP LAB/AI.PYTHON"
Even numbers: 4
```

Odd numbers: 4

Task Description #3 (AI-Based Code Completion for Class Attributes Validation)

Task: Use an AI tool to complete a Python class that validates user input.

Expected Output:

- AI-generated class with validation logic.

- Verification of condition handling.

- Test cases for valid and invalid inputs.

Prompt:

Generate a Python class User that validates age and email using conditional statements

CODE:-

```
class User:
    def __init__(self, name, age, email):
        self.name = name
        self.age = age
        self.email = email

    def validate_age(self):
        if self.age < 0:
            raise ValueError("Age cannot be negative.")
        elif self.age < 18:
            raise ValueError("User must be at least 18 years old.")
        else:
            return True

    def validate_email(self):
        if "@" not in self.email:
            raise ValueError("Email must contain '@'.")
        elif "." not in self.email:
            raise ValueError("Email must contain a valid domain.")
        elif self.email.startswith("@") or self.email.endswith("."):
            raise ValueError("Invalid email format.")
        else:
            return True

    def register(self):
        if self.validate_age() and self.validate_email():
            return f"User {self.name} registered successfully."
        else:
            return "Registration failed."
```

Example usage

```
user1 = User("Alice", 25, "alice@example.com")
print(user1.register())
```

OUTPUT:-

```
PS C:\Users\kisho_gtbim4w\OneDrive\WTMP LAB> & C:
\Users\kisho_gtbim4w\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/
Users/kisho_gtbim4w/OneDrive/WTMP LAB/AI.PYTHON"
User Alice registered successfully.
PS C:\Users\kisho_gtbim4w\OneDrive\WTMP LAB> & C:
\Users\kisho_gtbim4w\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/
Users/kisho_gtbim4w/OneDrive/WTMP LAB/AI.PYTHON"
User Alice registered successfully.
```

Task Description #4 (AI-Based Code Completion for Classes)

Task: Use an AI code completion tool to generate a Python class for managing student details.

Expected Output:

- AI-generated class code.

- Verification of correctness and completeness of class structure.

- Minor manual improvements (if needed) with justification.

Prompt:

Generate a Python class Student with attributes (name, roll number, marks) and methods to calculate total and average marks

CODE:-

```
def __init__(self, name, roll_number, marks):
    self.name = name
    self.roll_number = roll_number
    self.marks = marks

    def total_marks(self):
        return sum(self.marks)

    def average_marks(self):
        if not self.marks:
            return 0
        return sum(self.marks) / len(self.marks)

# Example usage
student = Student("Alice", 101, [90, 85, 88, 92])
print("Name:", student.name)
```

```
print("Roll Number:", student.roll_number)
print("Total Marks:", student.total_marks())
print("Average Marks:", student.average_marks())
```

OUTPUT:-

```
PS C:\Users\kisho_gtbim4w\OneDrive\WTMP LAB> & C:
\Users\kisho_gtbim4w\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/
Users/kisho_gtbim4w/OneDrive/WTMP LAB/AI.PYTHON"
Name: Alice
Roll Number: 101
Total Marks: 355
Average Marks: 88.75
```

Task Description 5 (AI-Assisted Code Completion Review)

Task: Use an AI tool to generate a complete Python program using classes, loops, and conditionals together.

Expected Output:

- Complete AI-generated program.
- Identification of strengths and limitations of AI suggestions.
- Reflection on how AI assisted coding productivity.

Prompt:

Generate a Python program for a simple bank account system using class, loops, and conditional statements

CODE:-

```
class BankAccount:
    def __init__(self, account_holder, balance=0):
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
        if amount > 0:
            self.balance += amount
            print(f"Deposited ${amount:.2f}. New balance: ${self.balance:.2f}")
        else:
            print("Deposit amount must be positive.")

    def withdraw(self, amount):
        if amount > 0:
            if amount <= self.balance:
                self.balance -= amount
                print(f"Withdrew ${amount:.2f}. New balance: ${self.balance:.2f}")
            else:
                print("Insufficient funds.")
```

```

        else:
            print("Withdrawal amount must be positive.")

    def check_balance(self):
        print(f"{self.account_holder}'s balance: ${self.balance:.2f}")

# Simple bank system
print("Welcome to the Bank System")
account = BankAccount("John Doe", 1000)

while True:
    print("\nChoose an option:")
    print("1. Deposit")
    print("2. Withdraw")
    print("3. Check Balance")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        amount = float(input("Enter deposit amount: "))
        account.deposit(amount)
    elif choice == "2":
        amount = float(input("Enter withdrawal amount: "))
        account.withdraw(amount)
    elif choice == "3":
        account.check_balance()
    elif choice == "4":
        print("Thank you for using the bank system.")
        break
    else:
        print("Invalid choice. Please try again.")

```

OUTPUT:-

```

PS C:\Users\kisho_gtbim4w\OneDrive\WTMP LAB> & C:
\Users\kisho_gtbim4w\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/
Users/kisho_gtbim4w/OneDrive/WTMP LAB/AI.PYTHON"
Welcome to the Bank System

```

```

Choose an option:
1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice: 1
Enter deposit amount: 10000
Deposited $10000.00. New balance: $11000.00

```

Choose an option:

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

Enter your choice: 3

John Doe's balance: \$11000.00

Choose an option:

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

Enter your choice: 2

Enter withdrawal amount: 100

Withdrew \$100.00. New balance: \$10900.00

Choose an option:

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

Enter your choice: